

Activity 4

*Dr. Ghada Fathy, Dr.
Dina Abdelhafize*

Part A: Q1: Predict output



```
#include <stdio.h>

int main() {
    int x = 0;
    int y = 5;
    printf("%d\n", x && y++);
    printf("%d", y);

    return 0;
}
```

Output

0

5

Q2: Predict output



```
int x = 5;

switch(x % 2)
{
    case 0: printf("Even");
    case 1: printf("Odd");
}
```

Output

Odd

Q3. Predict output



```
int i;  
  
for(i = 0; i < 5; i++)  
{  
    if(i == 2)  
        continue;  
  
    printf("%d ", i);  
}
```

Output
0 1 3 4

Q4. Predict output

```
int arr[2][2] = {{1,2},{3,4}};  
int i, j, sum = 0;  
  
for(i = 0; i < 2; i++)  
{  
    for(j = 0; j < 2; j++)  
    {  
        sum += arr[i][j];  
    }  
}  
  
printf("%d", sum);
```

Output

10



Q5. Predict output

```
int x = 0;

if(x == 0)
    printf("A");
if(x)
    printf("B");
else
    printf("C");
```

Output
AC



Part B: Q1. Correct the Code

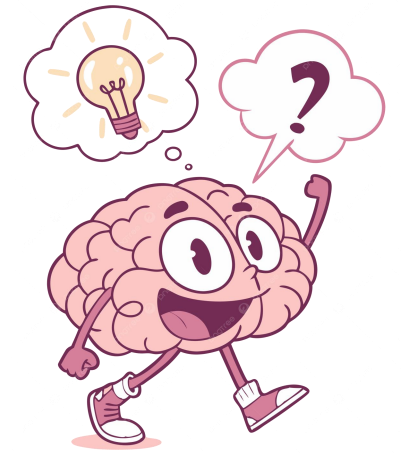


```
int x = 1;

switch(x)
{
    case 1: printf("A");
    case 2: printf("B");
}
```

```
switch(x)
{
    case 1: printf("A"); break;
    case 2: printf("B"); break;
}
```

Q2. Correct the Code



```
int i = 0;

while(i < 5)
{
    if(i == 2)
        continue;

    printf("%d ", i);
    i++;
}
```

X Infinity loop , i not incremented
when i == 2

```
while(i < 5)
{
    if(i == 2)
    {
        i++;
        continue;
    }

    printf("%d ", i);
    i++;
}
```


Q3. Correct the Code

```
float x = 2.0;  
  
switch(x)  
{  
    case 2: printf("Two");  
}
```

✗ Switch not working with fraction numbers

```
int x = 2;  
  
switch(x)  
{  
    case 2: printf("Two");  
}
```



Q4. Correct the Code

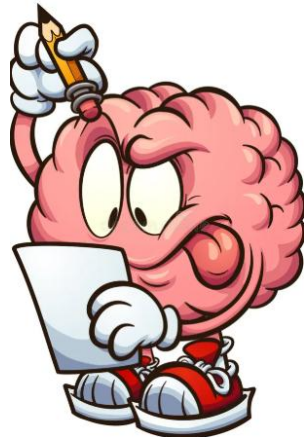
```
int i;  
  
for(i = 1; i > 5; i++)  
{  
    printf("%d ", i);  
}
```



X Condition false from start
Correct $i < 5$

BONUS Question

- Write a C program that:
 - Reads a single character,
 - a string (without spaces),
 - a sentence (must include spaces)
 - prints them in the same order on three separate lines



```
#include <stdio.h>
```

```
int main() {
```

```
    char ch;
```

```
    char word [10];
```

```
    char sentence [50];
```

```
    printf ("Please Enter the Character \n");
```

```
    scanf("%c",&ch);
```

```
    printf ("Please Enter the Word \n");
```

```
    scanf("%s",word);
```

```
    getchar();
```

```
    printf("Please Enter the Sentence \n");
```

```
    fgets(sentence,sizeof(sentence),stdin);
```

```
    printf("The Character is %c \n",ch);
```

```
    printf("The Word is %s \n",word);
```

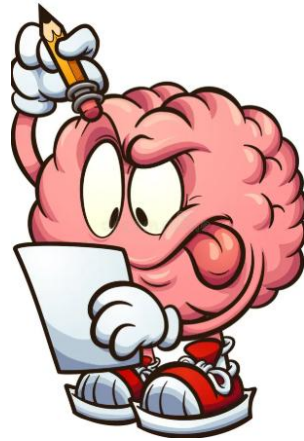
```
    printf("The sentence is %s \n", sentence);
```

```
    return 0;
```

```
}
```

BONUS Question

- Write a C program that:
 - Take integer n from user
 - Prints number from 1 to n using loop

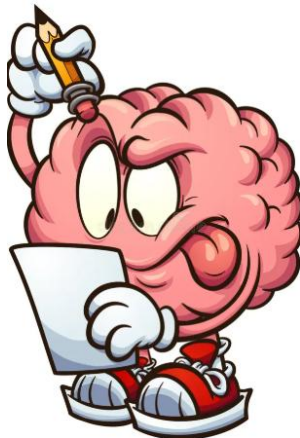


```
#include <stdio.h>
```

```
int main() {  
    int n=0;  
    printf("Enter the number\n");  
    scanf("%d",&n);  
    for (int i=1; i<=n; i++){  
        printf("%d\n" , i);  
    }  
    return 0;  
}
```

BONUS Question

- Write a C program that:
 - Take an integer from the user
 - Print whether it is even or odd



```
#include <stdio.h>
```

```
int main() {
```

```
    int input = 0;
```

```
    printf ("Please Enter the number \n");
```

```
    scanf("%d",&input);
```

```
    if (input%2 == 0){
```

```
        printf("Number is Even \n");
```

```
    } else{
```

```
        printf("Number is Odd \n");
```

```
    }
```

```
    return 0;
```

```
}
```

BONUS Question

- Write a C program that:
 - Take integer number from user.
 - Check:
 - If number is positive AND even → print "Positive Even"
 - Otherwise → print "Other"



```
#include <stdio.h>
```

```
int main() {
```

```
    int num = 0;
```

```
    printf ("Please Enter the number \n");
```

```
    scanf("%d",&num);
```

```
    if ( num >0 && num%2 == 0){
```

```
        printf("Positive Even \n");
```

```
    } else {
```

```
        printf("Other \n");
```

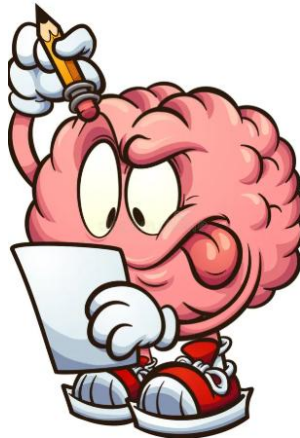
```
    }
```

```
    return 0;
```

```
}
```

BONUS Question

- Write a C program that:
 - Print numbers from 1 to 10 but:
 - Skip multiples of 3



```
#include <stdio.h>
```

```
int main() {
```

```
    for (int i=1; i<= 10; i++){
```

```
        if(i % 3 == 0)
```

```
            continue;
```

```
        printf("%d\n" , i);
```

```
    }
```

```
    return 0;
```

```
}
```